

PLAYO
VENUE OBSERVATORY

Teammate study guide and project walkthrough

Historical venue data · PostgreSQL/Supabase · SQL · machine learning ·
Cloudflare dashboard

Use this guide to understand the project, explain it in a viva, run the live demo and moderate public venue submissions.

Prepared for the project team · 2026

START HERE

How to use this guide

You do not need to read the source code first. Follow the guide in three passes:

1	Understand the story Read pages 3–12 to learn why the project exists, how the data was prepared and what the model means.
2	Practice the website Use pages 13–18 as a click-by-click walkthrough of the live dashboard and venue submission flow.
3	Prepare the viva Use pages 19–23 for deployment, demo order, troubleshooting and short answers to likely questions.
One-sentence project description We transformed a historical third-party Playo-related venue export into an auditable PostgreSQL/Supabase analytics system with SQL EDA, a leakage-aware rating model and a public Cloudflare dashboard.	

Live website

<https://playo-venue-observatory.dhivaa2004.workers.dev>

GitHub

<https://github.com/dhivaa-2004/playo-bdm-capstone>

PROJECT

The project in 90 seconds

The original material is a historical third-party Playo-related dataset. It is useful, but it is not a live Playo database. We preserve the raw files, remove only exact duplicates, normalize the content into PostgreSQL and keep source provenance on every historical venue.

SQL views provide the dashboard KPIs. A feature store creates region, coordinate and activity features. The target is avgRating. Entity-grouped splitting prevents the same venue group from entering both train and test. Predictions are written back to PostgreSQL and explained on the public dashboard.

Public venue submissions are a separate feature. They do not change historical KPIs or ML. They start as pending, are protected by Cloudflare Turnstile and become visible only after owner moderation.

Question	Short answer
What problem?	Make historical venue data queryable, reproducible and explainable.
What database?	PostgreSQL hosted by Supabase.
What frontend?	React/Vite dashboard hosted on Cloudflare Workers.
What ML task?	Estimate avgRating from non-leaking historical features.
Is it live Playo data?	No. It is a historical third-party snapshot; the current site was only a domain/schema reference.

SYSTEM

How the full system works

Layer	What it does	Why it matters
Raw archive	Preserves four regional JSON files unchanged.	The evidence can be re-audited.
Python ETL	Profiles, validates, deduplicates and writes load SQL.	Cleaning logic is reproducible.
Supabase	Stores normalized PostgreSQL tables, views and predictions.	The database is the source of dashboard truth.
SQL + ML	Runs EDA, builds feature_store_v1 and evaluates models.	Results are traceable to defined queries and features.
Cloudflare	Serves the React app and API routes.	The project is publicly accessible without ChatGPT hosting.

Critical separation

Historical records drive the research metrics and model. Community submissions stay in submitted_venues and never enter historical KPIs, model training or predictions.

End-to-end flow

Raw files → Python audit/ETL → normalized PostgreSQL → SQL views → feature store → grouped ML → predictions → Worker API → React dashboard

Dataset audit and the numbers to remember

Measure	Result	Meaning
Raw rows	3,701	Rows across the four source-region files.
Exact-deduplicated venues	3,697	Four exact duplicates removed.
Rated venues	3,186	86.2% have avgRating.
Unrated venues	511	Missing target, never a zero-star rating.
Activity/service labels	89	Not all labels are sports; some are services.
Source regions	4	Bangalore, Chennai, Delhi/NCR and Hyderabad.

Deduplicated regional counts

Bangalore	Hyderabad	Delhi / NCR	Chennai
1,358	1,085	698	556

Why no large synthetic expansion?

Synthetic rows would not repair unknown observation dates, geographic selection bias, sparse ratings or missing bookings/prices. The real historical dataset is sufficient for the primary empirical analysis.

EVIDENCE

How to describe the data honestly

Use this exact academic positioning:

Recommended wording

The project uses a historical third-party Playo-related dataset as its empirical base. The current Playo website was reviewed only as a domain/schema reference. The project does not claim that the records are current Playo listings.

Never claim:

- We scraped current Playo venues.
- All records are current and verified.
- A rating prediction measures demand, revenue or popularity.
- Missing ratings mean poor quality.

Provenance fields

Field	Historical value	Community value
source_type	third_party_historical	user_submitted
source_dataset	playo-find-venue-master	public_venue_submissions
is_synthetic	false	false

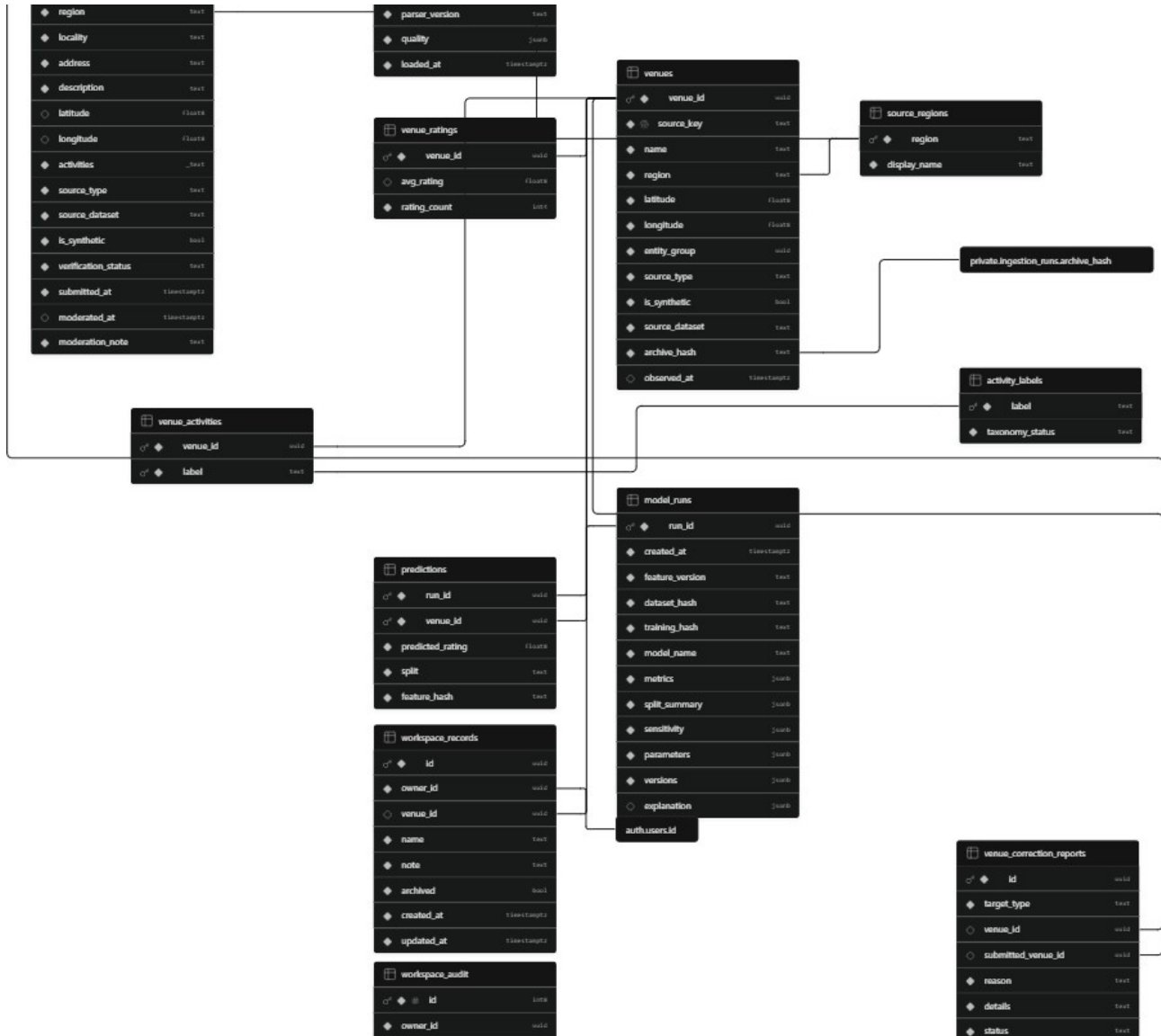
Database design in plain language

Table	Purpose	Important link
source_regions	Reference list of the four source regions.	venues.region → source_regions.region
venues	One row per historical venue with coordinates and provenance.	Primary key venue_id
venue_ratings	Average rating and rating count.	One-to-one with venues
activity_labels	Controlled list of activity/service labels.	Primary key label
venue_activities	Many-to-many bridge between venues and labels.	venue_id + label
model_runs	Model version, metrics, parameters and split summary.	Primary key run_id
predictions	Prediction per run and venue.	run_id + venue_id
submitted_venues	Moderated public venue submissions.	Separate from venues
venue_correction_reports	Public reports of possible data problems.	Targets historical or submitted venue

Normalization idea

Venue identity, ratings and activities are stored separately. This avoids repeating the same venue information for every activity and makes SQL joins explicit.

Implemented ER diagram



Supplied Supabase ER diagram, cropped only to remove unused white space.

How to explain it

- Start at venues in the middle: it is the historical entity table.
- source_regions and venue_ratings connect directly to venues.
- venue_activities connects venues to activity_labels, creating a many-to-many relationship.
- predictions connects one venue to one model_runs record for each run.
- submitted_venues and venue_correction_reports form a separate moderated path.

Python ETL: what happens to the raw files

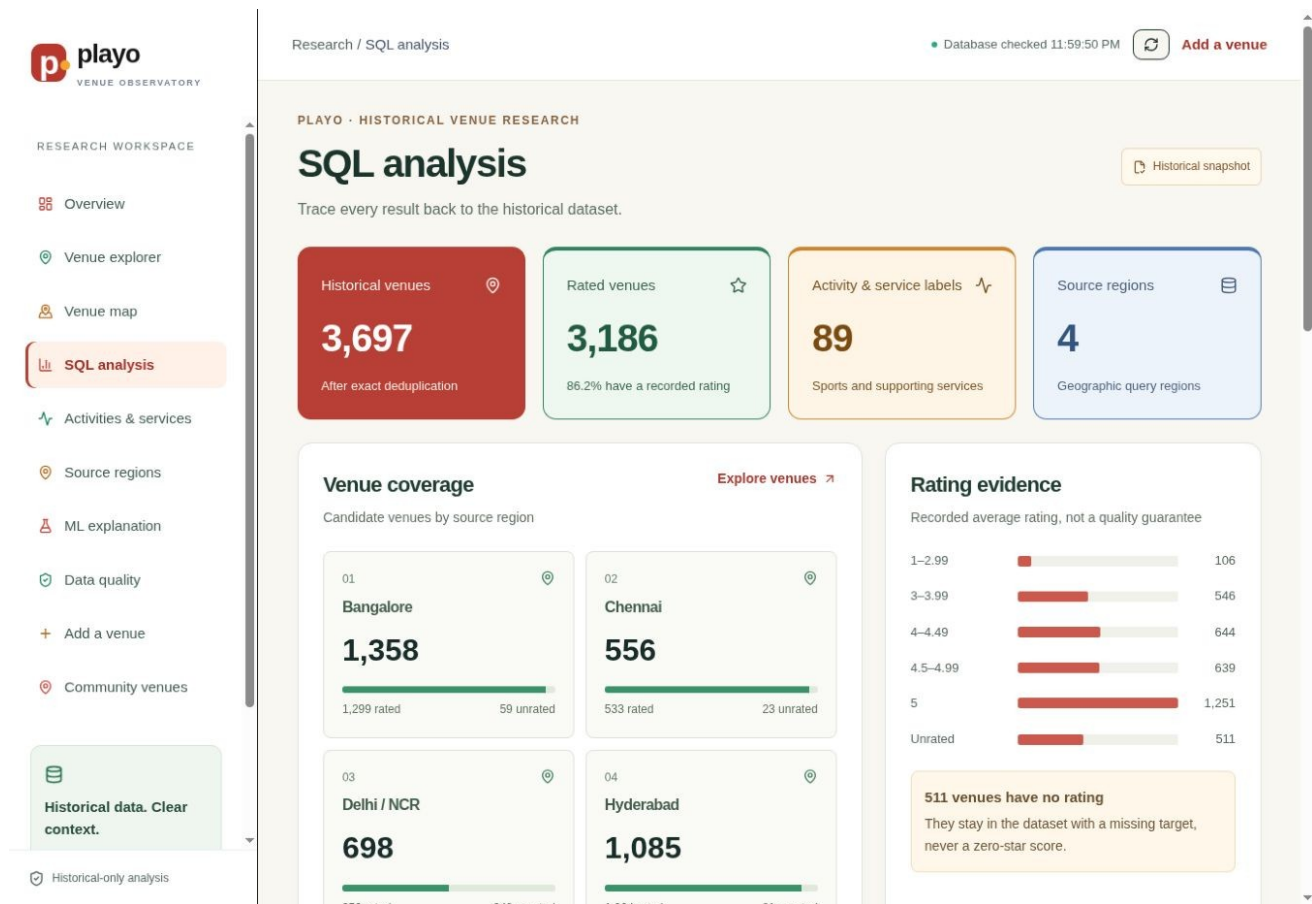
1	Archive Calculate a stable archive hash and retain the original files unchanged.
2	Profile Count rows, nulls, duplicates, cities/regions, labels, ratings and coordinates.
3	Normalize Convert values to defined types and split venues, ratings and activity labels into relational tables.
4	Deduplicate Remove exact duplicate records only; do not merge similar names without evidence.
5	Load Run the generated SQL in a transaction, then compare table counts with the quality report.

Quality rules to remember

- Coordinates must be valid latitude/longitude values.
- avgRating is either missing or between 1 and 5.
- rating_count cannot be negative.
- Historical records must keep source_type, source_dataset and archive_hash.
- Raw info HTML and phone information are not exposed in the public UI or ML feature store.

ANALYSIS

SQL analysis: what the dashboard is querying



SQL analysis page: database-backed counts, regional coverage and rating evidence.

Click-by-click

- 1 Open SQL analysis**
In the left sidebar, click SQL analysis.
- 2 Read the four KPIs**
Historical venues, rated venues, labels and source regions come from Supabase views.
- 3 Compare regions**
Scroll to regional coverage and note that source regions are query regions, not verified municipal boundaries.
- 4 Interpret missing ratings**
The unrated category contains missing targets; it is not a zero-star bucket.

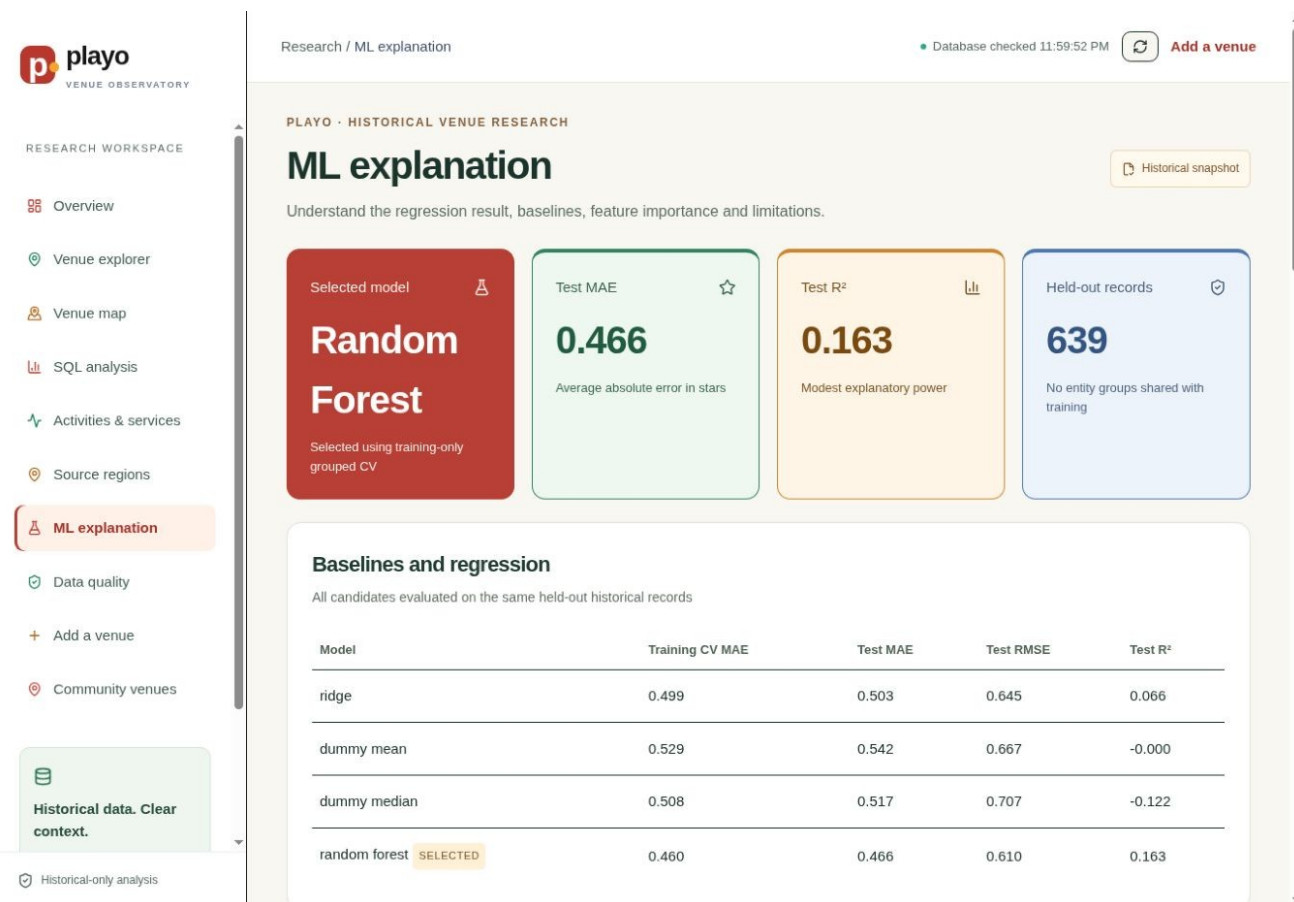
Feature store and leakage prevention

Use as a feature	Exclude from first model	Reason
Source region	rating	rating is a truncated copy of avgRating.
Latitude/longitude	info HTML	The HTML repeats rating and rating count.
Activity indicators	Name, links, icons, IDs	Identity-like fields encourage memorization.
entity_group for splitting	ratingCount	Used first as a reliability/sensitivity variable.
Target avgRating is the supervised regression target. Venues without avgRating are excluded from supervised evaluation, then receive an unrated prediction after the model is selected.		
Split rule The train/test split is grouped by entity_group. Result: zero entity-group overlap between train and held-out test data.		

Why this matters

Without leakage controls, the model could appear extremely accurate only because a feature already contains the answer or the same entity appears on both sides of the split.

How to explain the model results



ML explanation page showing selected model, held-out metrics and baselines.

Measure	Value	Plain meaning
Selected model	Random Forest	Chosen by grouped cross-validation.
Test venues	639	Unseen entity groups used only for final evaluation.
MAE	0.466	Average absolute error is about 0.47 rating stars.
RMSE	0.610	Larger errors receive more weight.
R²	0.163	The model explains a modest share of rating variation.
Median baseline MAE	0.517	The selected model improves on a simple constant prediction.

Correct conclusion

The model has useful research signal, but predictions are estimates—not guarantees of venue quality, demand, popularity or revenue.

Walkthrough 1: start at Overview

Research / Overview

Database checked 11:59:06 PM [Add a venue](#)

PLAYO · HISTORICAL VENUE RESEARCH

A clearer view of the venue landscape.

Explore the empirical dataset, understand its limits, and follow the evidence.

See the recorded venue landscape from every angle.

Explore 3,697 historical venue records, compare source regions and examine the evidence behind the ratings.

[Explore venues](#) [View SQL analysis](#)

Sports imagery for context; these are not photos of listed venues. Photos: CHUTTERSNAP and Anurag Sharma.

Historical venues	Rated venues	Activity & service labels	Source regions
3,697	3,186	89	4
After exact deduplication	86.2% have a recorded rating	Sports and supporting services	Geographic query regions

Venue coverage [Explore venues](#)

Candidate venues by source region

Rating evidence

Recorded average rating, not a quality guarantee

Overview page of the live Cloudflare-hosted dashboard.

1	Confirm live status The top-right status should change from Connecting to database to Database checked.
2	Read scope first The page says historical venue research and labels the content as a historical snapshot.
3	Use the KPI cards Quote 3,697 venues, 3,186 rated venues, 89 labels and four source regions.
4	Use the sidebar The sidebar is the recommended demo order; each label is a real route, not a decorative button.

Venue explorer: search and filter

The screenshot shows the 'Venue explorer' interface. On the left is a sidebar with navigation options: Overview, Venue explorer (selected), Venue map, SQL analysis, Activities & services, Source regions, ML explanation, Data quality, Add a venue, and Community venues. The main area is titled 'Venue explorer' and includes a search bar, filters for regions, activities, and ratings, and a table of results.

PLAYO · HISTORICAL VENUE RESEARCH

Venue explorer

Search the historical snapshot. These listings are not verified current availability.

Viewing historical records. [Browse user-submitted venues](#) →

Search venue names... **Search** All regions All activities & services Any rating count

Any average rating Name A-Z Reset

3,697 matching venues

Historical third-party data - 20 results per page

Venue	Source region	Activities / services	Rating	Rating count
040 Arena	Hyderabad	Basketball Box Cricket Football +1	☆ 5.00	8
1 Cube Sports & Gamez Hub	Chennai	Badminton Table Tennis	☆ 3.86	14
1 Cube sports & Gamez Hub - Guindy 2.0	Chennai	Badminton	☆ 4.00	4
1 Pickleball - Rooftop	Hyderabad	Pickleball	☆ 5.00	6
10 Sports Club - Saturn	Chennai	Box Cricket Football	☆ 5.00	4
10 Sports Club Badminton	Chennai	Badminton	☆ 5.00	2

Venue explorer with live filters and historical record rows.

1

Click Venue explorer

Use the left sidebar.

2

Search by name

Type a venue name and click Search.

3

Narrow the results

Choose a source region, activity/service, rating count or minimum average rating.

4

Open a venue

Click a venue name or the arrow at the right of a row to view its details and prediction context.

Interpretation rule

These are recorded historical listings. Do not describe the table as verified current availability.

Walkthrough 3: use the Venue map

The screenshot displays the Playo Venue map interface. On the left is a sidebar with navigation options: Overview, Venue explorer, **Venue map** (selected), SQL analysis, Activities & services, Source regions, ML explanation, Data quality, Add a venue, and Community venues. Below the sidebar is a button for 'Historical data. Clear context.' and a checkbox for 'Historical-only analysis'. The main content area is titled 'Research / Venue map' and includes a timestamp 'Database checked 12:04:24 AM' and an 'Add a venue' button. The section is labeled 'PLAYO · HISTORICAL VENUE RESEARCH' and 'Venue map'. A description states: 'Explore recorded coordinates without treating them as current venue availability.' Below this is an 'Interactive historical venue map' section with a dropdown for 'Source region' set to 'All regions'. The map itself shows India with numerous orange dots representing historical venue coordinates, primarily concentrated in the northern and central regions. A zoom control is visible in the top-left corner of the map.

Interactive Leaflet map with 3,697 historical coordinate records.

1

Click Venue map

Use the left sidebar.

2

Filter a source region

Use the drop-down at the top-right of the map.

3

Zoom and inspect clusters

Use +/-. Map points help explain regional concentration.

4

Keep non-map records available

The Venue explorer remains the complete list; the map is a geographic view, not the only access path.

Map disclaimer

Coordinates and source-region assignments come from the historical third-party dataset and may be outdated or imprecise.

Add a venue: public submission

No-sign-in public form. Every new submission is reviewed before public display.

1	Click Add a venue Use the sidebar or the top-right action.
2	Fill required fields Venue name, source region, locality/area and at least one activity are required.
3	Optional details Address, description and coordinates may be added when known.
4	Pass Turnstile and submit Cloudflare Turnstile reduces automated spam. The database inserts verification_status = pending.
What pending means The submission is saved in Supabase immediately, but it is private until an owner changes the status to approved.	

Owner moderation in Supabase

A public visitor cannot approve their own submission. The owner uses the Supabase dashboard:

1	Open the project Supabase Dashboard → dhivaa-2004's Capstone → Table Editor.
2	Open submitted_venues Use the public schema and select the submitted_venues table.
3	Find the pending row Check name, region, locality, activities, submitted_at and any duplicate warning.
4	Set verification_status Choose approved for public display or rejected when the claim should not be shown.
5	Record the decision Set moderated_at and add a short moderation_note. Supabase Table Editor auto-saves the edited cell; there may be no separate Save button.
6	Verify the public result Return to Community venues and click Refresh. Only approved rows should appear.
Important Approval means suitable for display after owner review. It does not mean Playo or real-world verification.	

Walkthrough 5: Community venues

Research / Community venues

Database checked 11:59:55 PM [Add a venue](#)

PLAYO · HISTORICAL VENUE RESEARCH

Community venues

Moderated community contributions, stored separately from the historical dataset.

[Community workflow](#)

Approved community venues

[Add a venue](#)

Owner-reviewed submissions; they remain unverified community claims.

Separate from historical research
Community entries never change historical KPIs, model training or predictions. Approval means suitable for display—it does not mean Playo or real-world verification.

Search approved venue names

[Search](#) [All regions](#) [Refresh](#)

No approved community venues match. New submissions remain private until reviewed.

Page 1 [Previous](#) [Next](#)

Independent academic capstone · Historical third-party Playo-related data [Methodology & source code](#)

Historical data. Clear context.

Historical-only analysis

Community page after demo cleanup: no approved rows are currently displayed.

1	Click Community venues Use the left sidebar.
2	Search approved rows Search by venue name and optionally select a source region.
3	Understand visibility Pending and rejected records are not returned by the public policy.
4	Keep the research separate Community entries do not change historical counts, SQL analysis, training data or predictions.

DEPLOYMENT

How GitHub, Cloudflare and Supabase work together

Service	Stores / runs	Team action
GitHub	Source code, migrations, documentation, tests and deliverables.	Push reviewed changes to main.
GitHub Actions	Frontend tests, build, deployment and health check.	Open Actions to inspect a failed run.
Cloudflare Workers	Public React assets and Worker API routes.	Check runtime variables and deployment logs.
Supabase	PostgreSQL tables, RLS policies, SQL views and model results.	Use Table Editor for moderation; SQL Editor for controlled queries.

Release flow

GitHub push → automated tests → Vite production build → Cloudflare deploy → production health check

Secrets and keys

- The browser uses only the Supabase publishable key.
- Never commit a service-role key, database password, API token or .env file.
- Deployment values belong in Cloudflare/GitHub environment settings.
- PostgreSQL row-level security—not secrecy of the publishable key—protects public database access.

A simple 7-minute live demo script

Time	Screen	What to say
0:00–0:45	Overview	Define the historical scope and quote the four KPIs.
0:45–1:45	Venue explorer	Search/filter records and open one venue.
1:45–2:30	Venue map	Show geographic clusters and the coordinate disclaimer.
2:30–3:30	SQL analysis	Explain that visible metrics come from database views.
3:30–5:00	ML explanation	Compare MAE with the baseline and explain modest R^2 .
5:00–6:15	Add a venue	Show the no-sign-in form and pending moderation flow.
6:15–7:00	Community venues	Explain approved-only RLS and historical separation.

If the database is slow

Pause until the top-right indicator says Database checked, then continue. Do not refresh repeatedly while the first request is still loading.

Suggested ending

The contribution is not a perfect prediction model. It is a transparent, database-backed workflow where data provenance, analysis, evaluation and public presentation remain connected.

Troubleshooting checklist

Symptom	Likely cause	What to check
Sidebar click appears inactive	Old cached deployment or route error.	Hard-refresh, confirm current Workers URL, inspect browser console and Worker health.
Connecting to database stays visible	Supabase request failed or timed out.	Cloudflare env vars, Supabase project status, RLS and API response.
Submitted row not visible publicly	It is pending/rejected or filtered out.	submitted_venues.verification_status must be approved; click Refresh.
Submission is not in Supabase	Turnstile/API validation failed.	Form error message, Worker logs and submission endpoint.
GitHub changed but site did not	Action failed or deployment used an older commit.	GitHub Actions run, build/deploy steps and current commit SHA.
Counts changed unexpectedly	Historical data may have been altered.	Archive hash, ETL quality report and historical table counts.

Do not fix production by editing historical rows manually

Historical corrections should be reproducible through ETL/migrations. Community moderation is the only routine Table Editor workflow.

Likely questions and short answers

1	Why PostgreSQL? The data has clear entities and many-to-many relationships, so constraints, joins and reproducible SQL are valuable.
2	Why Supabase? It provides managed PostgreSQL, RLS and a publishable client API while keeping the database central.
3	Why not use both rating and avgRating? rating is a truncated copy of the target, so it would leak the answer.
4	Why group the train/test split? To prevent related copies of the same venue entity from appearing in both sets.
5	Why not train on synthetic data? The historical sample was sufficient, and synthetic scale would not solve missing dates/features or selection bias.
6	What does $R^2 = 0.163$ mean? The model explains only a modest amount of rating variation; it is useful for demonstration, not a strong causal/business predictor.
7	How are public submissions protected? Turnstile, server validation, duplicate checks, pending status and RLS-approved visibility.
8	Is this an official Playo product? No. It is an independent academic capstone based on a historical third-party dataset.

Team checklist before submission or demo

1	Check 1 Open the live Workers URL and wait for Database checked.
2	Check 2 Confirm the four overview KPIs: 3,697 / 3,186 / 89 / 4.
3	Check 3 Search one venue and open its detail page.
4	Check 4 Open SQL analysis and ML explanation.
5	Check 5 Explain that the model's MAE is 0.466 and R^2 is modest at 0.163.
6	Check 6 Explain that community submissions start pending and remain separate.
7	Check 7 Confirm no demo/test venue is left approved after testing.
8	Check 8 Open the GitHub README and presentation files.
9	Check 9 Never expose service-role keys, database passwords or API tokens.
10	Ready Use the historical-snapshot disclaimer in every presentation.

Final message

The project is complete when the data, database, model, dashboard, documentation and deployment all tell the same honest story.